

“
Tonight, 54,000
people
are coming to my
show.

*And I have no idea what's going to
happen.*

— Rajan, Event Director · 22 years in live events

CROWDTWIN

Crowd Safety, Reimagined



THE STORY BEGINS

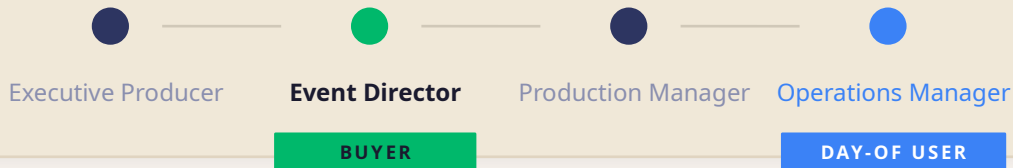


RAJAN

EVENT DIRECTOR · THE BUYER

The artist announcement changed everything.

WHERE RAJAN SITS IN THE ORGANISATION



Then the label called. The world's biggest pop star. One night only. Every ticket sold in 11 minutes. 54,000 confirmed.

"Suddenly I couldn't sleep. Because I'd seen what happens when a crowd this size goes wrong."

TODAY'S REALITY

This is how event organizers manage 54,000 people.

Rajan's full toolkit for the biggest night of his career:



Walkie-talkies & radio

Staff radio in crowd updates. Rajan waits for calls. By the time he hears about a problem, dozens are already trapped.



Manual headcounts

Staff at each gate hold a clicker. Numbers phoned to a central desk. Updated every 15 minutes. On paper.



CCTV cameras

A wall of 24 screens in a trailer. One security officer watching all of them. Impossible to monitor 54,000 people at once.



Experience and instinct

"We've done this before. It'll probably be fine." The professional hope that nothing goes catastrophically wrong.

THE PROBLEM

By the time Rajan knows — it's too late.

T - 0

Crowd surge begins
in Section 4

T + 7min

CCTV operator spots
it

T + 12min

Gates open. Crowd
redirected

T + 3min

Staff radio: "Getting
tight here"

T + 9min

Rajan is informed

T + 15min

3 people treated for
crush injuries



15 minutes is not a response time. It's a casualty report.

THIS IS NOT RARE

Rajan isn't the only one who's been here.

Astroworld Festival

10 deaths, 300+ injuries

Houston, Texas · 2021

50,000 attendees. No live crowd data. Staff overwhelmed. The crowd compressed for 40 minutes before the show was stopped.

Seoul Halloween Crowd Crush

159 deaths

Itaewon, South Korea · 2022

An alleyway became a fatal bottleneck. Witnesses called emergency services 11 times before anyone responded. No monitoring system existed.

Roskilde Festival

9 deaths in the crowd

Denmark · 2000

Crowd surge during Pearl Jam's set. Staff had no way to see the crush forming at the front. The band didn't know until it was over.

The common thread:

0

venues had live crowd density data

0

automated alerts fired

0

staff had a map of where to act

THE INSIGHT

Nobody could see where the crowd was.

In every single incident, organisers had no real-time picture of crowd density across their venue. They were flying blind.



Invisible crowd

Density builds silently. No one sees it.



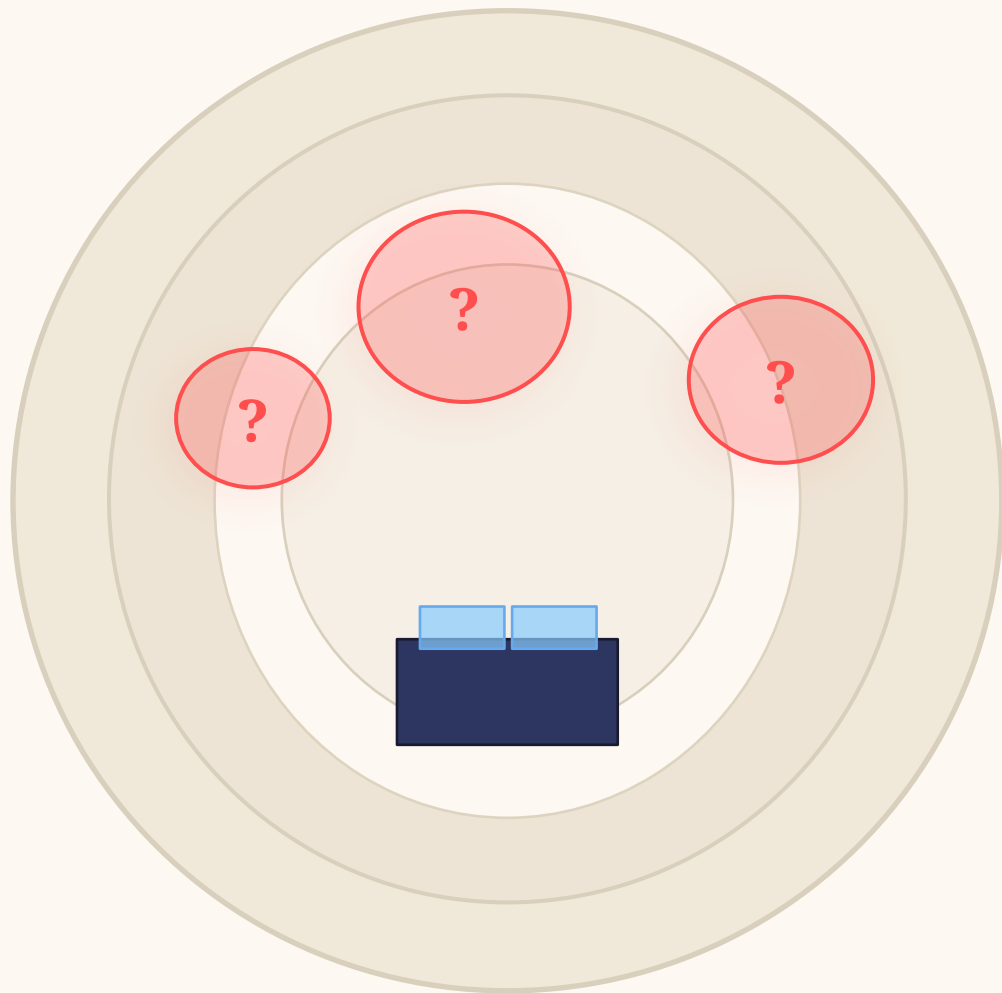
Too slow to react

Manual systems lag 10-15 minutes behind reality.



No map to act on

Staff don't know where to go or what to open.



The crowd isn't unpredictable.

It just hasn't been modelled.

Every crowd follows the same physics. The same bottleneck geometry. The same social pressure patterns. We can simulate all of it — before a single person arrives.

01

Crowds follow physics & behaviour rules

Individual agents move by density, bottleneck geometry, and social pressure. These patterns are simulatable — before the event even starts.

02

Venue layout is a predictable constraint

Every gate, corridor, and section creates known flow patterns. A digital twin of the venue encodes this structure permanently.

03

Deployment can be optimised in advance

If you know where pressure will build, you can pre-position forces. CrowdTwin tells Rajan where to send who — and when.

CrowdTwin

A Predictive Command Center powered by your venue's digital twin.

VENUE LAYER



01 Digital Twin

A live, graph-based model of your venue — every gate, zone, corridor, and exit. Built once. Updated continuously.

PREDICTION LAYER



02 Agent-Based Modelling

Simulates thousands of individual crowd agents using behavioural rules. Predicts surge zones, bottlenecks, and density thresholds before they form.

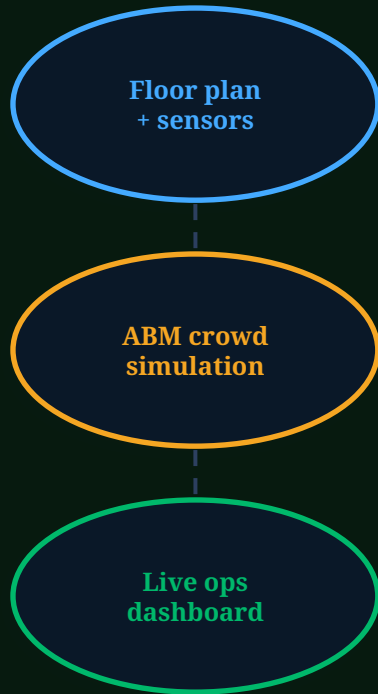
OPERATIONS LAYER



03 Command Center

Live dashboard for Rajan's team. Force deployment map, alert thresholds, one-tap redeployment orders, and full incident audit trail.

HOW IT CONNECTS



THE SOLUTION

CrowdTwin

A live crowd intelligence platform that gives event directors and operations managers complete visibility — before, during, and after every event.

WHO USES CROWDTWIN



Event Director

Plans, configures & buys

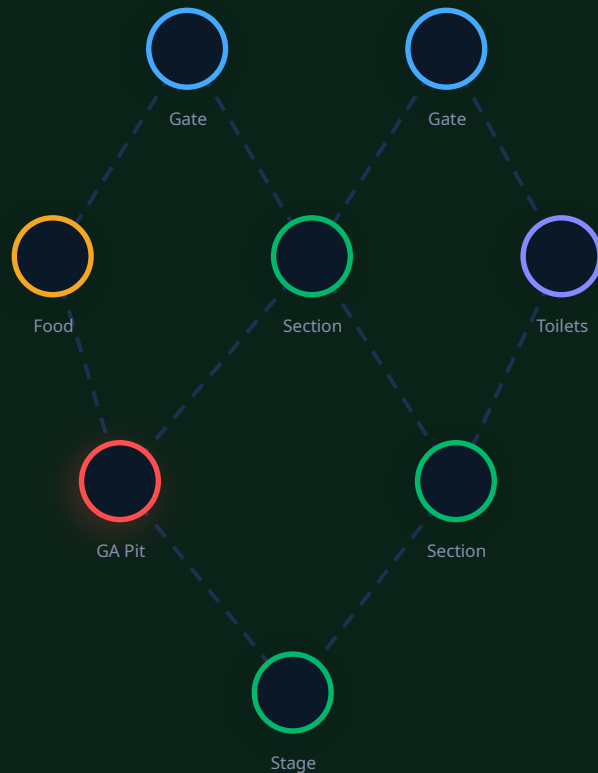


Operations Manager

Runs it live on the night

NODES = LOCATIONS

EDGES = PATHS



Three layers. One live picture.

01 VENUE LAYER

Digital Twin

A live graph of every gate, zone, corridor and exit — built once, updated continuously from your floor plan.

02 PREDICTION LAYER

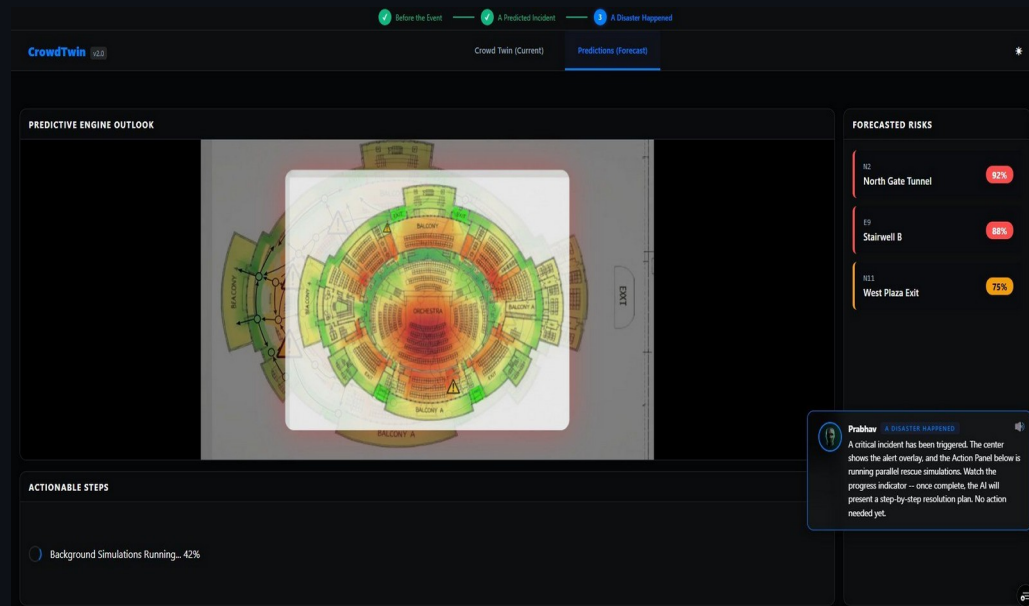
Agent-Based Modelling

Thousands of simulated crowd agents. Surge zones, bottlenecks and density thresholds predicted before they form.

03 OPERATIONS LAYER

Command Center

Live dashboard with one-tap redeployment orders, alert thresholds and a full incident audit trail.



Live from the CrowdTwin platform — venue heatmap, zone density & one-tap action panel

Rajan builds the plan before a single ticket is sold.

CROWDTWIN PLANNER · Wembley Stadium · June 14



WHAT RAJAN SETS UP IN ADVANCE



Entry & exit paths

Map every gate, turnstile and exit. Define safe capacity for each. CrowdTwin monitors them all from day one.



Crowd flow routes

Predict where crowds will move between gates, stands, and the floor. Set up steward positions along bottleneck paths.



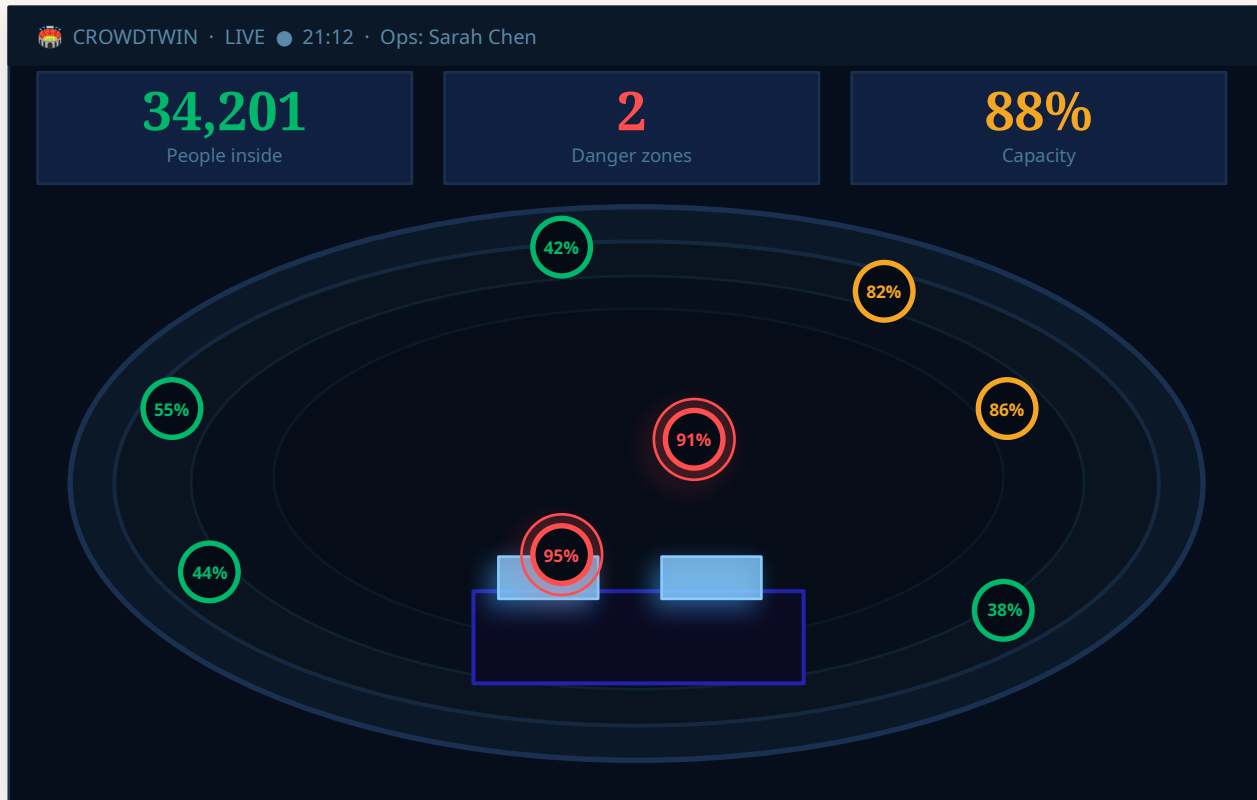
Commercial placement

Position food stalls, merch and bars exactly where crowd flow is highest. Higher dwell time = higher revenue.



BONUS — Commercial uplift

On the night, the Ops Manager has every zone at their fingertips.



CRITICAL

GA Pit — 91%

Enable single-file entry now

CRITICAL

South Stand — 95%

Seat limit reached

WARNING

Gate East — 82%

Redirect to Gate North

ONE-TAP ACTIONS

 Open Gate South

 PA redirect now

 Deploy stewards

One platform. Two phases. Complete control.

PRE-EVENT · RAJAN (EVENT DIRECTOR)

01



Map the venue

Plot gates, stands, exits and flow paths as a graph. Define capacity for every zone. Takes 30 minutes.

02



Place commercial points

Position food, bars, and merch based on predicted crowd flow. Optimise dwell time and spend per head.

LIVE EVENT · OPS MANAGER

03



Connect sensors

Turnstiles, cameras, Bluetooth beacons. All inputs merge into one live view. Auto-syncs with your floor plan.

04



See it live

Animated crowd density across every zone. The ops manager watches it all from one screen.

05



Act before it's bad

Alerts before zones hit capacity. Open a gate, PA redirect, deploy staff — all in one tap.

THE OUTCOME

The night of the show.

“21:14 — My screen shows Gate East at 78%. Yellow. I radio the team to pre-position stewards.”

“21:22 — Floor Front hits 88%. Orange. I open the overflow gate before anyone notices.”

“21:34 — Headliner walks on stage. 54,000 people. Every zone green or yellow.”

“21:35 — I watch the crowd flow in real time. For the first time in 22 years, I know exactly what’s happening. Everywhere. At once.”

What changed

15min

Average lead time to detect a crowd problem

0

Dangerous surges that went undetected

< 90s

Average response time with CrowdTwin

100%

Of the show passed without incident

What changes for Rajan's firm when CrowdTwin is running.

WITHOUT CROWDTWIN

Forces deployed on instinct and past experience

15+ minute lag from surge to response

Paper logs — impossible to reconstruct post-incident

No way to improve — every event starts from scratch



WITH CROWDTWIN

Optimal deployment plan generated by simulation before arrival

Alerts fire at 80% density — forces move before a surge forms

Full digital audit trail — one export for regulators & insurers

Each event refines the model — compounding safety dividend

Event organizers own the safety risk.

They hire the security firm — and carry the licence, liability and reputational risk if things go wrong. CrowdTwin gives them the oversight they can't get anywhere else.

MUSIC & FESTIVALS

Concert & Festival Promoters

80,000+

avg attendees — major festivals

Live Nation, AEG, independent promoters. PSARA-certified contractors are mandatory. One cancellation ends a career.

SPORTS & STADIA

Stadium & Arena Operators

£2.4M

avg liability per crush incident

Premier League clubs, T20 franchises, arena operators. SIA & safety officer requirements mandated by governing bodies.

CORPORATE EVENTS

Corporate & Exhibition Venues

3,200+

large corporate events yearly (India)

Venue managers, exhibition organisers, government events. Crowd incidents damage brand far beyond the event itself.

Event organizers don't just want safer events — they need evidence. CrowdTwin gives them the audit trail, the data, and the control.

Every crowd has a story.

CrowdTwin makes sure it has a safe ending.

CrowdTwin